Forcing-agent attribution: how the residual and its decomposition are computed

CESM2 diffusion emulator, checkpoint ep0860

Definitions

Let $s \in \{c, m\}$ index the side (CESM2 reference, emulator) and $X \in \{A, G, E\}$ the experiment: A all-forcing (`ssp370`), G GHG-only (`ghg`), E aerosol-only (`aaer`). The averaging window is $\mathcal{W} = [2041, 2050]$, chosen after 2015 so that no major volcanic eruption falls inside it, and before 2050, where the single-forcing runs end.

The evaluation files store per-member anomaly maps $x^s_{X,i}(t, \phi, \lambda)$, each referenced to its own side's 1850–1900 climatology, for member $i = 1, \ldots, N^s_X$.

Step 1 — decadal mean per member.

$$\bar{X}^s_i(\phi, \lambda) \;=\; \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} x^s_{X,i}(t, \phi, \lambda) \tag{1}$$

Step 2 — ensemble mean.

$$\langle X^s \rangle \;=\; \frac{1}{N^s_X} \sum_{i=1}^{N^s_X} \bar{X}^s_i \tag{2}$$

Under equal sampling (`-match-members`) the emulator ensemble is truncated to the reference count, $N^m_X = N^c_X$, so that neither side's mean is better converged than the other's. Here $N_A = 3$, $N_G = 10$, $N_E = 10$.

Step 3 — the residual field.

$$R^s \;=\; \langle A^s \rangle - \langle G^s \rangle - \langle E^s \rangle \tag{3}$$

R^s is what the two agents fail to explain between them. It is not a pure interaction term: the single-forcing runs hold ozone, land use, biomass burning, solar and volcanic forcing at 1850, so $R =$ (other forcings) + (nonlinear interaction).

Step 4 — the Arctic operator. With area weights $w(\phi) = \cos\phi$, renormalised inside the mask $\phi \geq 60°$N,

$$\langle\!\langle f \rangle\!\rangle \;=\; \frac{\displaystyle\sum_{\phi \geq 60°} \sum_{\lambda} w(\phi)\, f(\phi, \lambda)}{\displaystyle\sum_{\phi \geq 60°} \sum_{\lambda} w(\phi)} \tag{4}$$

The five bars

The two end bars are the Arctic-mean residual on each side,

$$b_1 = \langle\!\langle R^c \rangle\!\rangle = +1.2180 \text{ K}, \qquad b_5 = \langle\!\langle R^m \rangle\!\rangle = +2.1416 \text{ K}. \tag{5}$$

The three middle bars are the emulator's bias for one experiment, carrying the sign with which that experiment enters Eq. (3):

$$\Delta_A = +\langle\!\langle \langle A^m \rangle - \langle A^c \rangle \rangle\!\rangle = +(4.674 - 4.242) = +0.4328 \text{ K}, \tag{6}$$
$$\Delta_G = -\langle\!\langle \langle G^m \rangle - \langle G^c \rangle \rangle\!\rangle = -(4.021 - 4.257) = +0.2354 \text{ K}, \tag{7}$$
$$\Delta_E = -\langle\!\langle \langle E^m \rangle - \langle E^c \rangle \rangle\!\rangle = -(-1.488 + 1.233) = +0.2554 \text{ K}. \tag{8}$$

The sign flip is the substance of the panel: the emulator's GHG-only run is too *cool* and its aerosol-only run too *cold*, yet both terms are *subtracted* in Eq. (3), so both push the residual *up*. Three biases of different physical sign act on R in the same direction.

Arctic mean (K)	CESM2	emulator	bias
All-forcing (ssp370)	+4.242	+4.674	+0.433
GHG-only	+4.257	+4.021	−0.235
Aerosol-only	−1.233	−1.488	−0.255
Residual R	+1.218	+2.142	+0.924

Why the waterfall closes exactly

$$\langle\!\langle R^m \rangle\!\rangle - \langle\!\langle R^c \rangle\!\rangle \;=\; \Delta_A + \Delta_G + \Delta_E \tag{9}$$
$$1.2180 + 0.4328 + 0.2354 + 0.2554 \;=\; 2.1416$$

Equation (9) is an identity, not a fit. Every operator in the chain — the decadal mean, the ensemble mean, the combination $A - G - E$, and the area-weighted mean — is linear, so they commute and

$$\langle\!\langle R^m \rangle\!\rangle - \langle\!\langle R^c \rangle\!\rangle = \langle\!\langle (A^m - A^c) - (G^m - G^c) - (E^m - E^c) \rangle\!\rangle. \tag{10}$$

Verified numerically, the closure error is 1.2×10^{-10} K, i.e. floating point. The panel therefore does not test anything: it *attributes* an already-known 0.924 K gap to its three sources, and the finding is that the split is roughly even rather than one bad scenario.

Caveat. After matching, Δ_A rests on three members per side, all the CESM2 `ssp370` reference in these files provides. It is both the largest bar and the least certain, and it is the one that would move were the reference rewired to the ten held-out LENS2 members. Bar labels in the figure are rounded to two decimals, so reading them off gives 2.15 rather than 2.14; the underlying values close exactly.